An exponential lower bound for integration under uniform derivative bounds

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Abstract

We consider integration on the unit cube for smooth functions whose every partial derivative, including the function itself, is bounded by one. For every fixed absolute error tolerance $0 < \varepsilon < 1$, we prove that deterministic integration requires at least $(1-\varepsilon)(\frac{1}{2} + \sin\frac{1}{2})^{-d}$ function evaluations in dimension d. The bound includes adaptive sampling and nonlinear reconstruction. The proof uses quadratic forms of tensor products of unit vectors: a projection makes the function vanish at the samples, while its integral stays large and all derivative bounds are preserved.

1 The integration problem

Uniform control of derivatives gives strong local regularity, but the cost of integrating such functions in high dimension is a separate question. For each integer $d \geq 1$, let

$$F_d = \left\{ f \in C^\infty([0,1]^d; \mathbb{R}) : \|D^\alpha f\|_\infty \leq 1 \text{ for every } \alpha \in \mathbb{N}_0^d \right\}, \qquad I_d(f) = \int_{[0,1]^d} f(x)\, dx. \tag{1}$$

Here D^α denotes the mixed partial derivative with multi-index α, and the supremum is taken on the closed cube. In particular, $\alpha = 0$ is included. Let $n(\varepsilon, d)$ be the smallest number of exact function evaluations needed by a deterministic algorithm with worst-case absolute error at most ε on F_d. The sites may be chosen adaptively and the output may depend nonlinearly on the sampled values.

Wojtaszczyk [2, Theorem 2.1] proved that, for any fixed number of samples, the minimal integration error on this class tends to one as $d \to \infty$. The exponential dependence on dimension remained a distinct question. Krieg and Ullrich [1, Section 8.1, equation (8.5) and the discussion after Proposition 8.3] explicitly distinguish it from the known curse for uniform approximation on the same class. Novak and Pillichshammer [3, Section 5.2] give related integration lower bounds for finite-summability derivative norms and positive quadrature formulas; their discussion leaves the supremum derivative norm unresolved.

Positive-matrix and squared-kernel arguments give exponential integration lower bounds in tensor Hilbert spaces [4, 5]. Those Hilbert unit balls need not satisfy (1); the trigonometric example in [5, Section 3.1] has a sup-norm embedding constant $2^{d/2}$. Here we use the operator norm of a projection to bound a quadratic function directly. Choosing the vector frequency to be $1/2$ preserves every derivative bound, and a tensor moment calculation retains the absolute integration scale.

Theorem 1. *Set $\lambda = \frac{1}{2} + \sin(\frac{1}{2})$, with angles in radians. For every $d \geq 1$ and $n \in \mathbb{N}_0$, any deterministic algorithm using at most n exact function evaluations has worst-case absolute integration error on F_d at least*

$$\max\{0, 1 - n\lambda^d\}. \tag{2}$$

Consequently, for every $0 < \varepsilon < 1$ and $d \geq 1$,

$$n(\varepsilon, d) \geq (1 - \varepsilon)\lambda^{-d}. \tag{3}$$

This includes adaptive algorithms and nonlinear reconstruction, without any restriction on quadrature weights.

Since $\frac{1}{2} < \lambda < 1$, equation (3) gives the curse of dimensionality: one may take $\varepsilon_0 = 1/2$, $c = 1/2$ and $\rho = \lambda^{-1} > 1$, in every dimension.

2 Proof

We construct a bounded function that vanishes at any prescribed sample sites. Its values are squared distances from a subspace in a finite Euclidean space. The vector defining this distance has derivatives of exact norm 2^{-k} at order k, which offsets the factors in the product rule.

For $t \in \mathbb{R}$ and $x \in \mathbb{R}^d$, define

$$u(t) = \begin{pmatrix} \cos((t - 1/2)/2) \\ \sin((t - 1/2)/2) \end{pmatrix}, \qquad U_d(x) = \bigotimes_{j=1}^{d} u(x_j) \in \mathbb{R}^{2^d}. \tag{4}$$

All vector norms below are Euclidean. The tensor product in (4) has one coordinate for each choice of a sine or cosine component in every factor.

Lemma 1. *If A is a real symmetric $2^d \times 2^d$ matrix with $\|A\|_{\mathrm{op}} \leq 1$, then $f_A(x) = U_d(x)^T A U_d(x)$ belongs to F_d.*

Proof. Let $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Then $u^{(k)}(t) = 2^{-k} J^k u(t)$ and $\|u(t)\|_2 = 1$, so $\|u^{(k)}(t)\|_2 = 2^{-k}$ for all $k \geq 0$. Each coordinate derivative acts on its own tensor factor. The squared norm of a tensor product is the product of the squared factor norms; therefore

$$\|D^\beta U_d(x)\|_2 = 2^{-|\beta|}, \qquad |\beta| = \beta_1 + \cdots + \beta_d. \tag{5}$$

For every $\alpha \in \mathbb{N}_0^d$, the product rule gives

$$D^\alpha f_A(x) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (D^\beta U_d(x))^T A D^{\alpha - \beta} U_d(x), \qquad \binom{\alpha}{\beta} = \prod_{j=1}^{d} \binom{\alpha_j}{\beta_j},$$

where $\beta \leq \alpha$ means coordinatewise comparison. By Cauchy–Schwarz and (5),

$$|D^\alpha f_A(x)| \leq 2^{-|\alpha|} \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} = 2^{-|\alpha|} \prod_{j=1}^{d} 2^{\alpha_j} = 1.$$

The finite sums and products defining f_A are smooth on all of $\mathbb{R}^d$. The estimate includes order zero and every repeated or mixed derivative. □

Lemma 2. *For any $m \in \mathbb{N}_0$ and sites $x^1, \dots, x^m \in [0,1]^d$, there is a function $f \in F_d$ such that $0 \le f \le 1$, $f(x^j) = 0$ for every j, and*

$$I_d(f) \ge 1 - m\lambda^d. \tag{6}$$

Proof. Let P be the orthogonal projection onto $V = \operatorname{span}\{U_d(x^1), \dots, U_d(x^m)\}$ and let $Q = \mathrm{Id} - P$. For $m = 0$, take $P = 0$. The projection Q is a symmetric contraction, so Lemma 1 applies to

$$f(x) = U_d(x)^T Q U_d(x) = \|Q U_d(x)\|_2^2.$$

Thus $0 \le f \le 1$ and $f(x^j) = 0$.

To estimate its integral, compute the one-dimensional moment matrix

$$M = \int_0^1 u(t)u(t)^T \, dt = \begin{pmatrix} \lambda & 0 \\ 0 & 1-\lambda \end{pmatrix}. \tag{7}$$

The off-diagonal entry vanishes by oddness about $1/2$, and

$$\int_0^1 \cos^2((t-1/2)/2)\, dt = \frac{1}{2} + \frac{1}{2}\int_0^1 \cos(t-1/2)\, dt = \frac{1}{2} + \sin(1/2).$$

The sine-square entry is its complement. Since $0 < \sin(1/2) < 1/2$, $\frac{1}{2} < \lambda < 1$. Product integration of the continuous entries yields

$$M_d = \int_{[0,1]^d} U_d(x) U_d(x)^T \, dx = M^{\otimes d}.$$

The matrix M_d is diagonal, has trace one, and every diagonal entry is at most λ^d. Write $r = \dim V \le m$ and choose an orthonormal basis $v_1, \dots, v_r$ of V. Then $P = \sum_{i=1}^r v_i v_i^T$, and

$$\operatorname{tr}(PM_d) = \sum_{i=1}^r v_i^T M_d v_i \le r\lambda^d \le m\lambda^d.$$

It follows that $I_d(f) = \operatorname{tr}(QM_d) = 1 - \operatorname{tr}(PM_d) \ge 1 - m\lambda^d$. Repeated or dependent sample sites simply reduce r. □

Proof of Theorem 1. Run the algorithm on the zero function and record its $m \le n$ sample sites, including its stopping decision. Apply Lemma 2 to these sites. Both f and $-f$ belong to F_d and return zero at every recorded site. By induction over the sampling steps, the algorithm follows the same path on either input and produces the same output a. For $b = I_d(f) \ge 0$, the triangle inequality gives

$$\max\{|b-a|, |-b-a|\} \ge b \ge 1 - m\lambda^d \ge 1 - n\lambda^d.$$

This proves (2), since an error is nonnegative. If an algorithm has worst-case error at most ε, then $1 - n\lambda^d \le \varepsilon$, which is (3). □

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References

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